

Exp No: 5-ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

Objectives:

- To understand the interfacing of an Arduino Uno with a seven-segment display.
- To implement a manual electronic counter using a push button.
- To simulate the counter circuit using Proteus software.
- To verify the Arduino Embedded C program for counting from 0 to 9.

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
```

```
int delay_time=200;
```

```
void setup() {
```

```
//configure pin2 as an input and enable the internal pull-up resistor
```

```
pinMode(12, INPUT_PULLUP);
```

```
pinMode(1,OUTPUT);
```

```
pinMode(2,OUTPUT);
```

```
pinMode(3,OUTPUT);
```

```
pinMode(4,OUTPUT);
```

```
pinMode(5,OUTPUT);
```

```
pinMode(6,OUTPUT);
```

```
pinMode(7,OUTPUT);
```

```
}  
void loop() {  
  //read the pushbutton value into a variable  
  int sensorVal = digitalRead(12);  
  
  if (sensorVal == LOW) {  
    x0=true;  
  }  
  while(x0){  
    zero();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
      x1=true;  
      x0=false;  
    }  
  }  
  while(x1){  
    one();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
      x2=true;  
      x1=false;  
    }  
  }  
  while(x2){
```

```
two();  
sensorVal = digitalRead(12);  
if (sensorVal == LOW) {  
  x3=true;  
  x2=false;  
}  
}  
while(x3){  
  three();  
  sensorVal = digitalRead(12);  
  if (sensorVal == LOW) {  
    x4=true;  
    x3=false;  
  }  
}  
while(x4){  
  four();  
  sensorVal = digitalRead(12);  
  if (sensorVal == LOW) {  
    x5=true;  
    x4=false;  
  }  
}  
while(x5){  
  five();  
  sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {  
  x6=true;  
  
  x5=false;  
}  
}  
while(x6){  
  six();  
  sensorVal = digitalRead(12);  
  if (sensorVal == LOW) {  
    x7=true;  
    x6=false;  
  }  
}  
while(x7){  
  seven();  
  sensorVal = digitalRead(12);  
  if (sensorVal == LOW) {  
    x8=true;  
    x7=false;  
  }  
}  
while(x8){  
  eight();  
  sensorVal = digitalRead(12);  
  if (sensorVal == LOW) {
```

```
x9=true;
x8=false;
}
}
while(x9){
  nine();
  sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {
  x0=true;
  x9=false;
}
}

}
```

```
void zero()
{
  for(int i=1;i<7;i++)
  {
    digitalWrite(i,HIGH);
    digitalWrite(7,LOW);
  }
  delay(delay_time);
}
```

```
void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}
```

```
void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

```
void three()
{
```

```
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void four()  
{  
digitalWrite(1,LOW);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);
```

```
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void five()  
{  
digitalWrite(1,HIGH);
```

```
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void six()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void seven()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);
```



```
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
}
```

```
void eight()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void nine()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);
```

```

digitalWrite(5,LOW);

digitalWrite(6,HIGH);

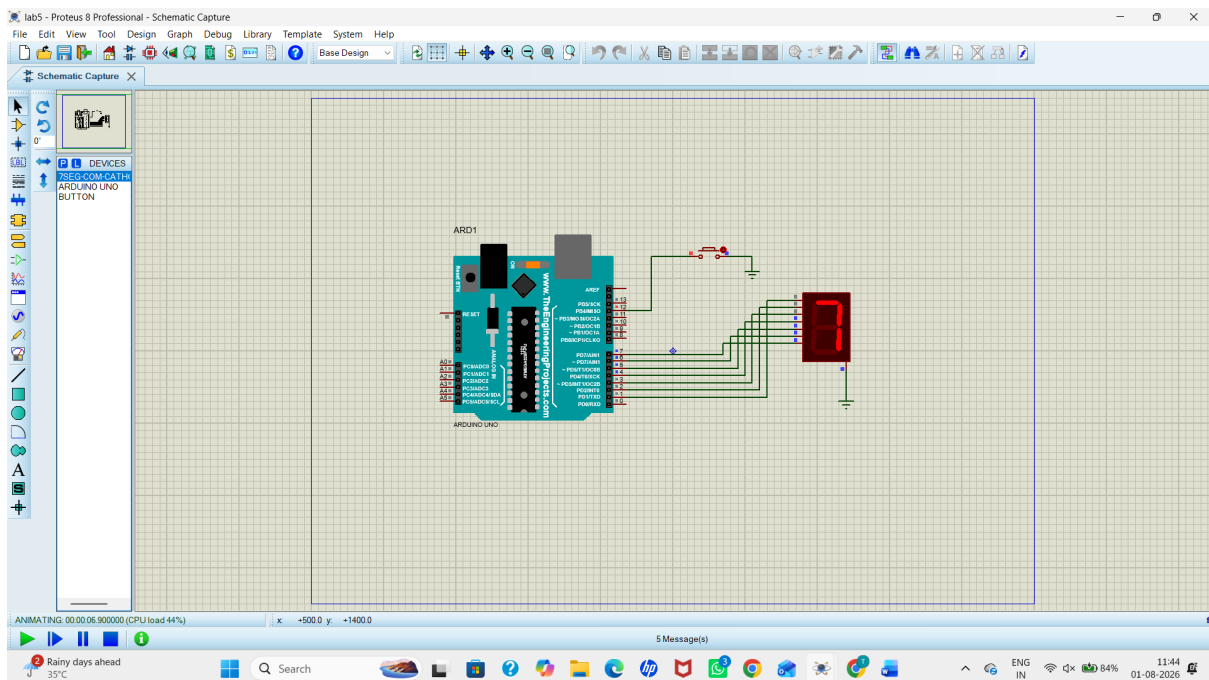
digitalWrite(7,HIGH);

delay(delay_time);

}

```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.